

Support-Two Normal Forms for a Shared-Tag Pauli Compilation Grammar

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Abstract

Tag-and-Restore compilation couples auxiliary anticommuting Pauli frames, a shared label operator, target assignments, and factored corrections. We analyze a fixed six-target, three-block grammar under a declared logical support-count objective. Every feasible solution can be transformed without increasing cost so that each auxiliary frame operator has support at most two. Support one is not uniformly sufficient: an exact two-qubit instance has unrestricted cost 5 and support-one cost 6. The sharp threshold yields a direct exact word-RAM algorithm with $O(n^9)$ time and $O(n^3)$ working memory for this fixed grammar. A standalone finite checker confirms the two local proof lemmas, while an implementation comparison agrees with a separate exact referee on the complete one-qubit domain and specified two-qubit stress cases. A pre-specified runtime comparison is adverse: the direct support-two implementation timed out on all six full-subject cells under a 120-second limit, although both solvers agreed on every jointly completed cell. The result is therefore a grammar-specific normal form and algorithmic upper bound, not evidence of faster compilation, hardware benefit, fault-tolerant resource advantage, or general block-encoding optimality.

Keywords: quantum compilation; Pauli operators; block encoding; exact optimization; normal forms; support complexity.

1 Introduction

Exact compilation is usually posed as optimization over a sufficiently expressive search space. A complementary question is whether that space admits a smaller exact normal form. Such a result can separate representational necessity from implementation performance and can expose the precise obstruction to a stronger restriction.

We study this question for a fixed Tag-and-Restore construction derived from a stabilizer-formalism block-encoding method [1]. An instance contains six nonidentity Pauli targets arranged into three ordered blocks. Each block uses an anticommuting auxiliary-frame pair, and the three blocks share one label operator. The declared objective counts auxiliary support and factored correction support with fixed coefficients. It is a logical structural coordinate, not a circuit-depth, fault-tolerant, or hardware cost model.

Main results and assumptions. All statements use exactly this six-target, three-block grammar and objective. First, we prove that every optimum has a representative in which each frame Pauli has support at most two. Second, an exact two-qubit counterexample proves that support one is not uniformly sufficient, so the uniform frame-support threshold is exactly two. Third, we count the ordered anticommuting support-two pairs and obtain a direct exact $O(n^9)$ word-RAM algorithm. Fourth, finite checks corroborate the local lemmas and exact implementation. Finally,

a pre-specified runtime study fails to show implementation benefit: the direct algorithm times out on all six full-subject comparisons.

The distinction between theorem and performance evidence is central. The normal form is an exact mathematical statement. The asymptotic result is an upper bound for a new direct algorithm, not a lower bound or a speed claim. The runtime study is retained as adverse evidence rather than converted into a positive result. No statement here implies physical-resource advantage or optimality for another grammar, objective, or compiler.

2 Grammar and objective

2.1 Pauli representation and block structure

Global phases are irrelevant to support counting. At each qubit a Pauli letter belongs to $\{I, X, Y, Z\}$, and the binary symplectic product determines commutation. The six targets are supplied as three block pairs. Each block selects an ordered anticommuting frame pair (A_0, A_1) , assigns its two targets to the two branches, and marks one branch as central. A shared Pauli S must realize the same two nonzero symplectic labels against every block's ordered frame pair. If six unpaired targets are supplied instead, the 15 possible pairings can be searched as a constant-size outer family.

For a target letter p and assigned frame letter f , the local correction letter is the phase-ignored product pf . Corrections in corresponding branches of the three blocks are factored together. For local letters a, b, c , the factor cost is

$$F(a, b, c) = w(a) + w(b) + w(c) - 2 \mathbf{1}[a = b = c \neq I],$$

where $w(I) = 0$ and $w(X) = w(Y) = w(Z) = 1$. Thus three identical nonidentity correction letters receive a two-unit sharing discount.

2.2 Declared objective

The total cost is coordinate-separable. Removing one nonidentity coordinate from a central frame refunds two units; removing one from a noncentral frame refunds four. The shared operator costs two units per supported qubit. The remaining term is the sum of F over both branches and all coordinates. Within-block target order, central choices, and the two common-label orientations form constant-size families because the grammar always contains three two-target blocks.

The unrestricted exact referee searches the complete admitted grammar. The support- k family restricts every frame Pauli to support at most k while retaining all other choices. Family containment immediately makes the support restriction an upper-bound candidate; equality requires proof or exact comparison.

2.3 Evidence boundary

The mathematical proof uses the objective above, a parity-preserving exchange, and a well-founded support-reduction argument. Finite enumeration checks the local lemmas but is not the logical basis of the all-size theorem. The exact counterexample is a lower witness for support two. Implementation comparisons establish conformance only on their stated domains. Runtime measurements address one implementation and panel, not the theorem's correctness.

3 Normal form, sharpness, and direct algorithm

3.1 A local correction bound

Lemma 1 (Correction exchange). *Fix one local target letter p , a nonidentity frame letter f , and the other two letters u, v in the same three-way factor. Replacing pf by p increases the factor cost by at most two:*

$$F(p, u, v) - F(pf, u, v) \leq 2.$$

Proof. Only one letter changes, so the ordinary nonidentity count rises by at most one. If the old triple did not receive the sharing discount, a new discount can only reduce the change. If the old triple did receive it, then pf was nonidentity and equal to $u = v$. Destroying the discount adds two, but the changed letter cannot increase the nonidentity count: either $p = I$ and the count falls by one, or p is another nonidentity letter and the count is unchanged. Hence the increase is at most two. $\square$

3.2 A parity-preserving subset

For one frame A , its anticommuting partner A' , and the shared operator S , assign each supported coordinate q the class

$$c_q = (\langle A_q, A'_q \rangle, \langle S_q, A_q \rangle) \in \mathbb{F}_2^2.$$

Global anticommutation implies that the first components sum to one.

Lemma 2 (Zero-sum subset). *If $w(A) \geq 3$, there is a nonempty proper subset $Q \subset \text{supp}(A)$ with $|Q| \leq 2$ and $\sum_{q \in Q} c_q = (0, 0)$.*

Proof. A zero class supplies a singleton. Otherwise, a repeated nonzero class supplies an equal pair, whose sum is zero. If neither occurs, all classes are distinct and nonzero. There are only three nonzero classes, so support at least three forces the support-three multiset containing each nonzero class once. Its first components sum to zero, contradicting global anticommutation. $\square$

3.3 Exact support threshold

Theorem 1 (Support-two normal form). *For every admitted instance and every qubit count, the unrestricted optimum equals the support-two optimum.*

Proof. Start from an unrestricted optimum. If a frame has support at least three, choose Q from the zero-sum lemma and replace its letters on Q by identities. The first parity condition preserves anticommutation with the partner, and the second preserves the shared-label syndrome. Because Q is proper, the modified frame remains nonidentity. At each removed coordinate the frame refund is at least two, while the correction-exchange lemma bounds the correction increase by two; the shared operator is unchanged. Cost therefore does not increase, and total frame support strictly decreases. Iteration terminates at an optimum in the support-two family. The reverse inequality follows from family containment. $\square$

The threshold is sharp. Write two-qubit Pauli strings with the qubit-0 letter first and take the three target pairs

$$(XI, XI), \quad (XI, XI), \quad (IX, IY).$$

A feasible support-two solution uses frame pairs (YI, XI) , (YI, XI) , and (ZX, IY) , marks the first branch central in each block, and takes $S = XI$ with common-label orientation $(1, 0)$. Its frame,

shared-operator, and correction costs are respectively 2, 2, and 1, for total cost 5. The support-one lower bound is exhaustive: on two qubits there are 12 ordered anticommuting support-one pairs per block, and the search covers their Cartesian cube, all four relative target orders, all eight central choices, both label orientations, and every compatible two-qubit shared operator. Its minimum is 6. A separate exact referee also returns 5, and independent reconstruction checks commutation, corrections, cost, and Pauli phases. Thus the uniform support threshold is exactly two.

3.4 Constructive pair count and complexity

There are $3n + 9\binom{n}{2}$ nonidentity phase-ignored Paulis of support one or two. For a weight-one first frame, the number of anticommuting support-two partners is $6n - 4$. For a weight-two first frame, splitting partners into weight one on the active support, weight two on the same support, and weight two with one shared coordinate gives $12n - 16$. The number of ordered anticommuting pairs is therefore

$$N_{\text{pair}}(n) = 3n(6n - 4) + 9\binom{n}{2}(12n - 16) = 54n^3 - 108n^2 + 60n.$$

One block pair uses at most three coordinates, so three pairs use at most nine. A minimum-cost shared operator has no support outside this active union, since outside letters change no constraint and only add cost. Its six symplectic equations can therefore be solved by a constant-size 64-syndrome dynamic program. After one linear scan of target coordinates, correction costs for each candidate are updated on at most nine active coordinates. Enumerating three ordered frame pairs yields

$$T(n) = O(n + N_{\text{pair}}(n)^3) = O(n^9)$$

on a word RAM, with $O(n + N_{\text{pair}}(n)) = O(n^3)$ working memory. Charging index and accumulated-cost bit operations adds polylogarithmic factors. This is an upper bound only; exponent optimality is not claimed.

3.5 Finite verification and adverse runtime result

A standalone implementation checks all 192 local correction cases and all odd-parity class tuples through support eight. It finds a maximum factor-cost increase of two, no subset-lemma failure above support two, and exactly four ordered support-two obstruction patterns. A separate implementation comparison verifies the constructive pair count through six qubits, the complete one-qubit target domain, specified two-qubit stress cases, active-union confinement of the shared operator, exact costs, witnesses, and phase reconstruction.

The implementation-performance study is adverse. Of 120 attempts specified before measurement, 108 completed and 12 timed out. Both exact solvers agreed on every jointly completed correctness cell, and every completed witness verified. The unrestricted referee completed all six full-subject cells. The direct support-two implementation timed out on all six corresponding cells under the 120-second limit. Consequently no runtime or memory improvement is established.

4 Discussion

The theorem separates representation from performance. It removes arbitrary-support frames from exact optimization without asserting that a direct enumeration is practically efficient. The adverse runtime study illustrates why this distinction matters: a polynomial exponent and a smaller normal form do not automatically outperform a factorized exact referee on finite workloads.

Table 1: Evidence layers and their authorized interpretation.

Evidence	Authorized conclusion
Analytic exchange proof	Exact support-two normal form for the declared grammar and objective
Exact cost $5 < 6$ witness	Support one is not uniformly sufficient
Constructive count	Direct $O(n^9)$ word-RAM upper bound
Finite implementation checks	Conformance on the checked domains
120-attempt runtime study	No measured production value; twelve timeouts retained

The sharpness witness also identifies the obstruction to support one. At support two, no nonempty proper subset need preserve both frame anticommutation and the shared-label syndrome. A locally removable frame letter can therefore be globally coupled to the shared operator. The upper proof and lower witness meet at this parity boundary rather than at an empirical cutoff.

The result is useful primarily as a formalization of one cost model. It supplies a complete representational reduction, an exact algorithmic upper bound, and a reproducible negative performance result. It does not establish a new universal synthesis primitive. Practical significance would require either a better algorithm exploiting the normal form, a broader theorem, or an independently evaluated resource improvement under a physically meaningful model.

5 Related work and contribution boundary

The studied grammar descends from Tag-and-Restore block encoding, which owns the primitive, freely chosen anticommuting frames, shared-label construction, and associated resource tradeoffs [1]. Anticommuting-unitary partitioning, Pauli-cluster compilation, and Pauli-level compiler representations are also established [2, 3, 4, 5, 6].

Support reduction itself is not new as a general idea. Phase-gadget synthesis decomposes Pauli exponentials using one- and two-qubit components [7], and parity-network work exploits sharing across related terms [8]. Two-qubit gates are universal for quantum computation [9], while stabilizer-state normal forms are classical tools of the Pauli formalism [10]. These results remove any claim that support-two building blocks, parity sharing, or two-local universality originate here. They address different objects from the grammar-specific statement proved above: the present theorem transforms auxiliary frames while preserving one fixed shared-label objective and exact optimum. Hamiltonian weight reduction and perturbative gadgets likewise change the encoded Hamiltonian, often by adding degrees of freedom or approximation, rather than proving this cost-monotone frame exchange [11, 12].

Current bounded searches also inspected Pauli compilers, exact and approximate synthesis, low-ancilla block encodings, fault-tolerant resource models, and compiler benchmarks. No exact equivalent of the full grammar-specific theorem was located, but a bounded search cannot establish absence or priority. The residual contribution is therefore stated descriptively: a cost-monotone normal-form proof, its sharp obstruction, and the induced exact upper bound for this declared grammar. Novelty beyond that narrow formulation remains a matter for independent expert review.

6 Limitations

The theorem is scoped to six targets, three blocks, one shared binary label, the stated factor rule, and fixed support multipliers. Changing the objective can make support three beneficial. Other term counts, shared-label ranks, block grammars, routing constraints, or factor rules require new proofs.

The direct complexity result is a word-RAM upper bound. It is not a lower bound, an exponent-optimality result, or an asymptotic statement about other implementations. The available finite verifier shares some algebraic primitives with the direct solver and is not an externally independent implementation of every witness identity.

The runtime panel is small and adverse. It provides no acceleration, memory, energy, logical-depth, fault-tolerant, or hardware claim. Earlier finite chemistry observations and heuristic structural-cost reductions are not used as theorem or production-value evidence here. Patents, private manuscripts, unindexed software, post-search publications, and independent venue-level novelty judgments were not assessable in the bounded literature audit.

7 Conclusion

For the declared shared-Tag Pauli compilation grammar, every optimum has a support-two frame representation, and an exact two-qubit witness shows that support one is insufficient. The sharp normal form yields a direct exact $O(n^9)$ word-RAM algorithm. Finite checks corroborate the proof and implementation, but a pre-specified runtime comparison finds no production advantage because the direct implementation times out on every full-subject cell. The durable result is therefore an exact, objective-scoped normal form with a preserved negative performance outcome, not a claim of general compiler or physical-resource superiority.

8 Data and code availability

The proof notes, standalone finite checker, direct solver, exact-comparison summary, sharpness witness, runtime specification, aggregate observations, and adverse result are available in the accompanying anonymous review archive. The archive includes integrity and verification instructions while excluding author-linked development history from the manuscript-facing surface. Protected data are not required for any claim in this paper.

Reproduction should proceed in three layers: rerun the standalone local-lemma checker; verify the exact-comparison summary and sharpness witness; then inspect the pre-specified runtime aggregate without changing its time limit, panel, or success rule. The twelve timeout rows are part of the result and must not be discarded.

9 Ethics, author contributions, and resource statement

This work uses no human participants, animals, clinical data, or personal data. Its principal integrity risks are overgeneralizing a grammar-specific theorem and suppressing negative performance evidence; the manuscript addresses both by stating the exact scope and retaining every timeout.

The author is responsible for the mathematical claims, source verification, code, and final manuscript. Large language models were used for literature-search assistance, proof-audit prompts, code and prose refinement, and manuscript formatting. All mathematical statements, citations,

calculations, and release decisions were checked against the cited sources and underlying evidence by the author.

No energy or carbon estimate is claimed. The exact finite checks and runtime study used ordinary CPU resources; the available records do not support a reliable life-cycle resource comparison.

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